



Harish Aravinth KM

B.Tech
Chemical Engineering
Indian Institute Of Technology, Tirupati

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CH)	Indian Institute of Technology, Tirupati	8.57(till 5th sem)	2023 - 2027
AISSCE	Aristo Public School, Cuddalore, TN	88.6%	2022
AISSE	Aristo Public School, Cuddalore, TN	87.2%	2020

PROJECTS

- **Target Drug delivery applications** May 2025 - Present
Advanced Drug Delivery and Smart Biomaterial Engineering, IIT Tirupati
 - Fabricated chitosan-based nanocarriers using ionotropic gelation and emulsion techniques; systematically optimized electrospinning parameters for biomedical-grade nanofibers.
 - Engineering injectable, fast-acting hemostatic hydrogels, focusing on gelation kinetics, mechanical stability, and clot-inducing performance under physiological conditions.
 - Developing bacteria-templated anisotropic cerium oxide nanomaterial with tunable stimuli-responsive (pH/NIR) behavior.
 - Designing bio-templated metal oxide architectures integrated with metal–organic framework (MOF) hybrids for advanced functional applications.
- **Linde Ammonia Synthesis Process** Aug 2024 - Dec 2025
Applied Process Engineering Course, IIT Tirupati
 - Completed process synthesis and flow-sheet development of the Linde ammonia process. Performed material and energy balance calculations, reactor–separator integration, and operating condition analysis under industrial constraints. Evaluated process efficiency, recycle optimization, and thermodynamic feasibility of ammonia synthesis loops.
- **Aspen Plus Process Simulation and Modeling**
Self-paced (NPTEL - YouTube)
 - Built and converged complete steady-state flowsheets in Aspen Plus for representative chemical processes. Modeled and analyzed key unit operations (reactors, distillation columns, pumps, heaters, valves) with appropriate thermodynamic property methods. Performed design evaluation and process optimization using simulation outputs and sensitivity analysis.
- **Chemical Engineering Web Platform (Frontend Development)** Jun 2025 - present
Built responsive website using HTML, CSS, JavaScript, ReactJS , IIT Tirupati [Github](#)
 - Developing a department-level web portal for Chemical Engineering at IIT Tirupati. Features include academic curriculum, laboratory details, faculty profiles, student resources, and research highlights.

EXPERIENCE

- **Process Engineer Intern** Jul 2025 - Jul 2025
Crimsun Organics Pvt. Ltd., Cuddalore Onsite
 - Assisted in operation, calibration, and monitoring of industrial process equipment. Supported plant troubleshooting, safety compliance, and efficiency improvement activities.
- **India Space Lab** Dec 2024 - Jan 2025
Intern Of Space Technology and Innovation Program Remote
 - Learned and trained in drone technology, CubeSat, astronomy, and space entrepreneurship.

TECHNICAL SKILLS

- **Programming & Simulation:** C, Python, Java, JavaScript, SQL, Matlab, AutoCAD, OriginPro, Aspen Plus
- **Web Development:** HTML, CSS, JavaScript, ReactJS
- **Libraries & Tools:** Pandas, Numpy, scikit-learn
- **OS & Tools:** Windows, Linux, Git & Github

AREA OF INTEREST

- Stimuli-responsive and targeted drug delivery systems
- Biomaterial-based nanoparticles and anisotropic nanostructures.
- Process modeling, simulation, and optimization using Aspen Plus
- Scientific and engineering-focused web platforms

KEY COURSES TAKEN

- Chemical Process and Calculation, Fluid and Particle Mechanics, Heat Transfer, Mass Transfer, Thermodynamics, Computational Techniques for Chemical Engineering, Numerical Methods, Chemical reaction engineering, Process dynamics and Control, Transport Phenomena, Separation Process, Introduction to NanoScience and Technology, Drug delivery-Principles and Application, Electrochemical Energy Storage, Introduction to Renewable Energy, Process Synthesis and Economics, Chemical Technology, Bioprocess Engineering, Introduction to Microfluidics.

POSITIONS OF RESPONSIBILITY

- **Vice-Captain**, Knights House Team, IIT Tirupati *Mar 2025 - May 2025*
- **Athletics IIT Tirupati Team Member**, Inter-IIT Sports Meets (IIT Gandhinagar(2023), IIT Indore(2024))
- **Co-coordinator**, Event Management Team , Chemical Engineering Club, IIT Tirupati *Jul 2025 - Present*
- **Sports Council Member**, Events and Management Team, IIT Tirupati *May 2025 - Present*
- **Core Member**, Designing and Content Writing Team, Rasoi Club, IIT Tirupati *May 2025 - Present*

ACHIEVEMENTS

- **Grandmaster** Abacus – SIP Academy, India *2020*
- **First Place** Poster Presentation - 'Alchemia' , IIT Tirupati *Apr 2025*
- **Second Place** Matlab Process Simulink - 'D-Solve' , IIT Tirupati *Nov 2025*
- **Two Gold Medals** Intra-IIT Sports Meet, Athletics Events, IIT Tirupati *2024*
- **Bronze Medal** Intra-IIT Sports Meet, Athletics Events, IIT Tirupati *2025*